

$$\zeta(-2n+1) \in \mathbb{Q}$$

Prove that $\zeta(-2n+1)$ are rational numbers for any positive integer n .

Proof: It is well known that $\zeta(-2n+1) = -\frac{B_{2n}}{2n} \in \mathbb{Q}$:

By definition $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$, where $\xi(s) = \xi(1-s)$. We show that $\zeta(2k) = \frac{(-1)^{k+1} (2\pi)^{2k} B_{2k}}{2(2k)!}$, then

$$\pi^{-k} \Gamma(k) \zeta(2k) = \xi(2k) = \xi(1-2k) = \pi^{k-1/2} \Gamma\left(-k + \frac{1}{2}\right) \zeta(1-2k),$$

where $\Gamma(k) = (k-1)!$ and $\Gamma\left(-k + \frac{1}{2}\right) = \frac{(-2)^k \sqrt{\pi}}{(2k-1)!!}$. Therefore

$$\zeta(1-2k) = \zeta(2k) \cdot \frac{\pi^{-2k} (k-1)! (2k-1)!!}{(-2)^k} = -\frac{B_{2k}}{2k} \in \mathbb{Q}.$$

By definition

$$\frac{z}{e^z - 1} = \sum_{n=1}^{\infty} B_n \frac{z^n}{n!} \text{ for } |z| < 2\pi.$$

Recall that

$$\pi \cot \pi z = \sum_{n \in \mathbb{Z}} \frac{1}{n+z} = \pi \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} = \frac{1}{iz} \cdot \left(\frac{\pi iz}{e^{\pi iz} - 1} + \frac{\pi iz}{2} \right) = \frac{1}{z} + \sum_{n=1}^{\infty} (-1)^n \frac{(2\pi)^{2n} B_{2n}}{(2n)!} z^{2n-1}$$

and

$$\sum_{n \in \mathbb{Z}} \frac{1}{n+z} = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2} = \frac{1}{z} + 2z \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{z^{2k}}{n^{2k+2}} = \frac{1}{z} - \sum_{k=1}^{\infty} 2\zeta(2k) z^{2k-1} \quad \forall |z| < 1.$$

Therefore $\zeta(2n) = (-1)^{n+1} \frac{B_{2n} (2\pi)^{2n}}{2(2n)!}$, and hence $\zeta(-2n+1) = -\frac{B_{2n}}{2n}$.

Since $\sum_{k=0}^n \binom{n+1}{k} B_k = 0$, $B_n \in \mathbb{Q}$ and $\zeta(-2n+1) \in \mathbb{Q}$.

Kummer's Congruence

Theorem: Let $k \in \mathbb{Z}_{\geq 1}$, $n_1, n_2 \geq k$ integers such that $n_1 \equiv n_2 \pmod{(p-1)p^{k-1}}$ and that $p-1 \nmid n_1$ (so $1 - n_1 \not\equiv 1$), then $\frac{B_{n_1}}{n_1} \equiv \frac{B_{n_2}}{n_2} \pmod{p^k}$, or equivalently $\zeta(1-n_1) \equiv \zeta(1-n_2) \pmod{p^k}$.

Without $n_1, n_2 \geq k$, we only have $\zeta^{(p)}(1-n_1) \equiv \zeta^{(p)}(1-n_2) \pmod{p^k}$ where $\zeta^{(p)}(s) = (1-p^{-s})\zeta(s) = \prod_{q \neq p} (1-q^{-s})^{-1}$ is the p -adic zeta function. So $-n \mapsto \zeta(-n)$ is p -adic continuous at negative integers.